Predictive Analytics - Course Project

1. Executive Summary

This report documents a five-week project for Amundi's retail engagement division to maximize expected profit from a weekly outreach campaign targeting up to 1,200 clients from a pool of 7,700 prospects. The approach evolved from descriptive business rules (Period 1), through Binary Classification (Period 2) and Regression-based investment estimation (Period 3), to a Combined Expected-Profit Model (Period 4) and finally to a fully calibrated Combined Model (Period 5).

The final strategy pairs a calibrated Random Forest classifier, selected for its superior evaluation metrics, with a Lasso regressor trained exclusively on confirmed investors. Together, the results of these models generate an Expected Profit formula that ranks every prospect by anticipated profit, ensuring our limited budget is directed at the highest-value opportunities.

2. Business Context

Amundi, Europe's largest asset manager and a subsidiary of Crédit Agricole, serves over 100 million retail clients across 35 countries. The retail engagement team operates within the bank-partnership channel, proactively identifying account holders likely to invest in UCITS-based funds. Each week, the team receives approximately 7,700 leads but can contact no more than 1,200, roughly 15.6% of available clients.

When a client invests, Amundi earns a 4.5% front-end commission on the amount invested. Each outreach attempt costs €5 regardless of outcome. This creates a highly asymmetric cost structure: a missed high-balance investor is far more costly than a wasted call on a non-investor. The core business objective is therefore to rank clients by expected profit and select the top 1,200. The Expected Profit formula used throughout is:

$$\mathbf{EP = P(invest) \times E(investment\ amount) \times 4.5\% - €5}$$

3. Data Overview & Preprocessing

The training data combined the baseline dataset (period_0.csv, ~7,700 records) with outcome data from Periods 1-4, yielding 12,229 records across 13 variables covering demographics, financial status, and marketing history. The key challenge is a strong class imbalance: only 11.4% of clients invested, which directly shaped model design decisions, including stratified sampling and the evaluation of ROC-AUC, Recall, and, once calibrated, Log-Loss as important metrics. A binary conversion indicator

was derived to support the classification stage, while a separate investor-only subset was maintained for regression, eliminating the noise of the large mass of zero investments.

Preprocessing was implemented within scikit-learn Pipelines to prevent data leakage: categorical variables were imputed with a constant 'missing' category and one-hot encoded with rare-category aggregation (min. frequency 0.10); numerical features were imputed with the median and standardized for the case of KNN, for instance. Missing values were moderate (e.g. education ~3.9%) and were handled consistently across all periods.

4. Classification: Predicting Investment Probability

4.1 Model Selection

Seven classifiers were evaluated using 5-fold stratified cross-validation with ROC-AUC as the primary metric: Logistic Regression, KNN, Ridge Classifier, Lasso Classifier, Random Forest, HistGradientBoosting, and XGBoost. KNN (AUC 0.622) and Ridge Classifier (AUC 0.579) were disregarded as they were not much better than a random model. The remaining models performed competitively on AUC (Fig.1).

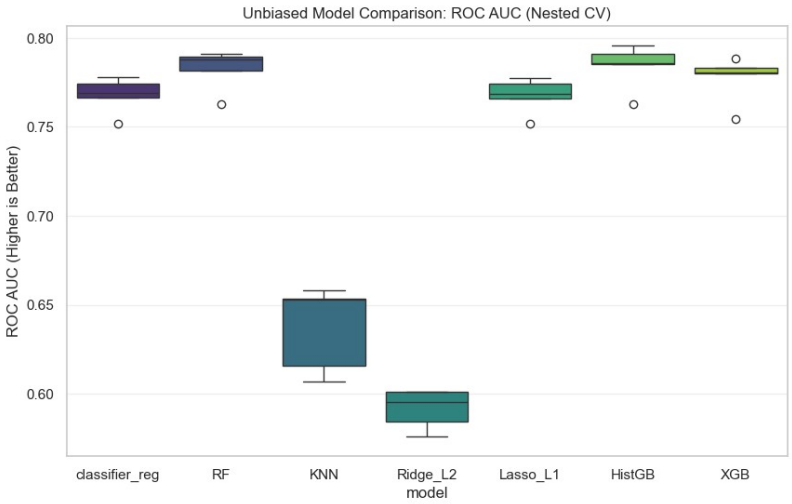


Figure 1. Comparison of the AUC across all models

The business cost structure, where missing an investor is far more damaging than a wasted call, made Recall an important criterion. As well as other metrics derived from the Confusion Matrix, Precision and F1, which we opted to follow with the 0.5 threshold, were evaluated across models, and the two finalists are compared in Table 1 below.

Model	ROC-AUC	Precision	Recall	F1
Random Forest	**0.794**	**38.8%**	**69.1%**	49.7%
HistGB	0.792	63.2%	28.7%	39.5%

Table 1. Finalist model comparison - Random Forest vs. HistGradientBoosting on the held-out test set.

HistGradientBoosting is very precise, but its recall of only 28.7% means it misses over 70% of actual investors, a commercially unacceptable false-negative rate given the asymmetric cost structure. Random Forest captures 69.1% of investors with a competitive AUC of 0.794 and a modest training-to-test gap of ~1.7 percentage points, confirming strong generalization. At this stage, Random Forest proves to be the better classifier, but we further investigate these two models.

4.2 Discriminative Power: Lift Analysis

The practical utility of the classifier, its ability to identify real investors in the ranked list, is confirmed by the Cumulative Response and Lift curves below. Targeting the top 16% of the ranked list (equivalent to 1,200 calls against 7,700 prospects) captures most available investors at a lift exceeding 3×, confirming that the weekly call budget is both sufficient and well-placed within the high-value segment, as shown in Fig. 2.

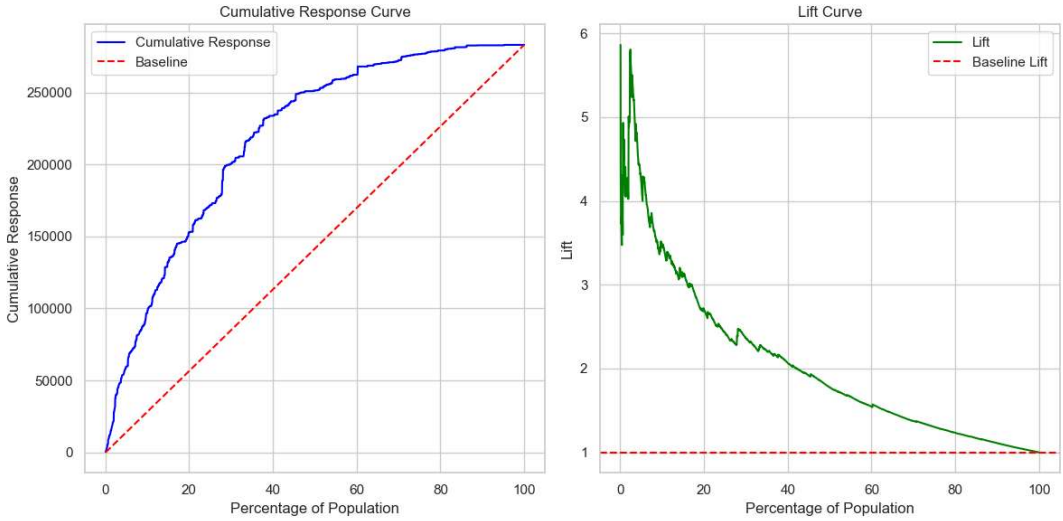


Figure 2. Cumulative Response Curve (left) and Lift Curve (right)

4.3 Probability Calibration

Raw Random Forest probabilities are systematically overconfident and feed directly into the EP formula, meaning inflated scores corrupt lead ranking. Probability calibration via Platt Scaling is therefore a necessary step before any EP calculation can be considered valid. After calibration, the Random Forest achieves a LogLoss of 0.367, which is lower than HistGB's calibrated LogLoss of 0.374. While both models retain almost identical ROC-AUC scores (Table 2). This confirms the Calibrated Random Forest as the classification model since it ranks leads as well as HistGB and produces more reliable probability to calculate the EP.

Model	Uncal. LogLoss	Cal. LogLoss	Cal. AUC
Random Forest	0.524	0.367	0.791
HistGB	0.370	0.374	0.789

Table 2. Calibration results - before and after Platt Scaling.

5. Regression: Predicting Investment Amount

5.1 Training Strategy

Training the OLS model on the entire dataset, including investors and non-investors, forced it to explain why most clients invest nothing rather than focusing on what drives investment volume. This results in a low R^2=0.24, because the dataset contains a strong class imbalance. Restricting the training data to confirmed investors, combined with regularization and interaction terms, substantially improved model fit across all evaluated approaches. This allows the regression to focus on estimating investment volume and identifying its key drivers, rather than being distorted by the dominant mass of zero outcomes.

5.2 Model Comparison & Champion Selection

Six regression models were evaluated on the investor-only subset, as summarized in Table 3. Linear regularized models (Lasso, Ridge, ElasticNet) dominate at $R^2 \approx 0.94$ and RMSE ≈ €162-163. Tree-based methods were substantially weaker (RF: $R^2 = 0.812$, RMSE = €318). Among confirmed investors, account balance is by far the strongest predictor of investment volume, with a Pearson correlation of 0.86 confirming a clear and direct linear relationship. This is precisely why regularized linear models outperform tree-based methods. The investor subset is relatively small, which leaves the tree models little room to leverage their ability to capture complexity without overfitting.

Model	RMSE (€)	R^2	Training Set
Lasso	**€161.66**	**0.942**	**Investors Only**
Ridge	€163.13	0.941	Investors Only
ElasticNet	€163.17	0.941	Investors Only
Random Forest Regressor.	€318.40	0.812	Investors Only
HistGB Regressor	€328.17	0.797	Investors Only
OLS Baseline	€347.46	0.240	Full dataset

Table 3. Regression model comparison - 5-fold cross-validated RMSE and R^2 on the investor-only training subset.

Lasso was selected as the best regression model. It produced the best RMSE at €161.66, and its L1 penalty reduces overfitting by shrinking irrelevant coefficients to exactly zero, effectively performing automatic feature selection, preventing the model from capturing noise in the interaction terms, and improving generalization to unseen data. The optimal regularization strength was found at a low alpha of 0.001, which is a meaningful result in itself: it indicates that, within the investor-only sub-population, multicollinearity is minimal and the relevant features carry a clear, strong signal. The model required only light penalization to generalize well, confirming that the investor-only subset is a highly structured and well-behaved training environment for linear regularized models.

The most influential retained predictors were all interaction terms with account balance or account balance itself, confirming its role as the dominant driver of investment volume. Account balance alone had the strongest overall effect (coefficient: 428.4), directly reflecting that wealthier clients invest larger amounts. The interaction between balance and tertiary education added a further positive effect (coefficient: 205.8), suggesting that highly educated clients with larger balances invest disproportionately more. The interaction between balance and retirement status showed the strongest negative effect (-198.9), indicating that retired clients with high balances are significantly less likely to invest large amounts.

6. Combined Model

Finally, we combine the classification model and the regression model into a single profit-driven ranking system. Rather than applying the two models sequentially, our strategy integrates them multiplicatively: for every client, the calibrated investment probability and the predicted investment amount are combined into a single expected profit: EP = P(invest) × E(investment amount) × 4.5% − €5.

The Calibrated Random Forest assigns each client a calibrated probability of investment P(invest), reflecting realistic probabilities rather than overconfident raw scores. The lasso regressor independently estimates an E(investment amount) for all clients, not just predicted investors, and the conversion probability acts as a weight that scales the potential return accordingly. Ranking all prospects by this EP score and selecting those with a positive expected profit ensures that every contacted client is backed by a clear commercial justification.

This architecture explicitly captures the interaction between willingness and wealth: a client with high conversion probability but a modest balance may rank below a moderate-probability client with a very high balance. Targeting by EP rather than by probability alone is what differentiates the Period 5 strategy from the pure classification approach used in Period 2.

7. Period 5 Final Submission

The combined Model pipeline was applied to the Period 5 dataset (6,711 records, approximately 13% fewer than prior periods). A distributional check confirmed no major drift in numerical features: age and balance distributions remained stable. The main deviation was in categorical variables; the education distribution shifted toward secondary-educated clients (~44% historically vs ~53% in P5), with tertiary-educated clients declining (~45% to ~30%). Since tertiary education was identified as a strong positive predictor of conversion, this shift was noted as a mild downside risk to predicted conversion rates.

In practice, the average calibrated probability across the selected leads was approximately 30.6%, which is roughly 2.7× the population base rate of 11.4%. All 885 selected contacts had a positive expected profit, leaving the remaining capacity unused rather than deploying it on unprofitable leads. The expected profit comes to 5,990€ with an average expected profit per call of 6.77€. Since the dataset contains fewer records than last week, the decline in profitable leads decreased from 929 to 885.

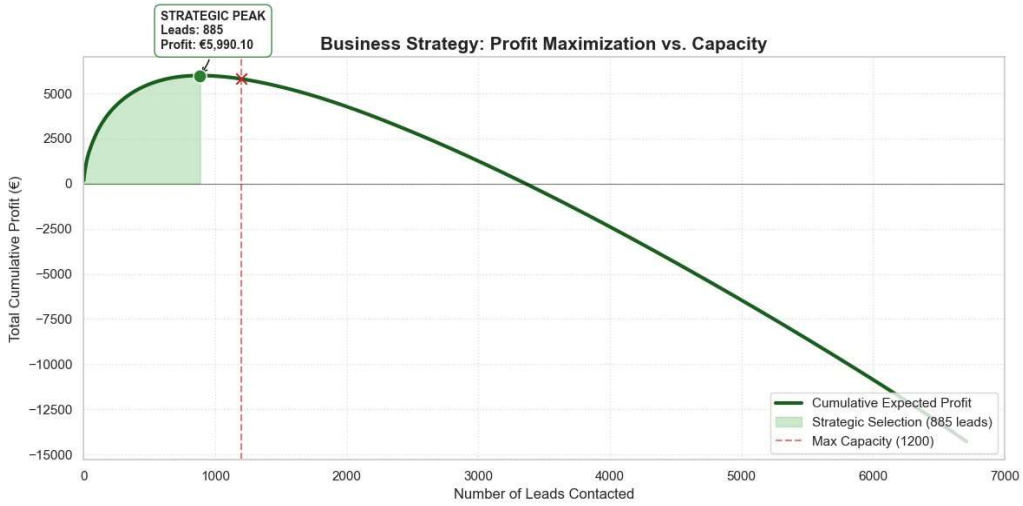


Figure 2. Cumulative Expected Profit curve for Period 5.

Following the campaign, the actual outcomes for the 885 contacted clients in period 5 were recorded and compared against the model's predictions. The campaign generated a total realized profit of €4,534, compared to the model's predicted €5,990. The deviation of approximately 24% is partially explained by the distributional shift in Period 5, where the decline in tertiary-educated clients reduced the share of the segment strongly associated with high conversion rates. Nevertheless, the model achieved a prediction accuracy of 94.4% in the final 4, confirming the robustness of the Combined EP Model's out-of-sample performance. The results confirm that a data-driven, model-based approach substantially outperforms random or rules-based outreach. Three recommendations follow for future implementation:

First, the combined EP Model should be retained as the standard framework, because targeting by conversion probability alone ignores wealth, while targeting by predicted investment amount alone ignores conversion risk. The combined model ranking captures both dimensions.

Second, probability calibration is mandatory before probabilities enter the EP formula. Uncalibrated scores distort lead ranking and must be treated as a non-negotiable step in any deployment pipeline.

Third, in datasets with strong class imbalance, where the majority of clients do not invest, training the regression model exclusively on confirmed investors allows the model to focus on the true drivers of investment volume, rather than being distorted by the dominant mass of zero outcomes.